

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 225 — Past Questions & Materials
by Renance

Course: MTH 225
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — MTH225 PAST QUESTIONS 2020/2021

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

SECOND SEMESTER EXAMINATION 2020/2021

Course Code/Title: MTH 225-Real Analysis II

Instructions: Attempt any 4 questions. Time Allowed: 2½ Hours.

Question 1 (17 ½ Marks)

(a) Define clearly Pointwise convergence of sequence of rvfs (3 marks)

(b) For each integer $n \geq 1$, define $f_n(x) = x^n - x^{n+1}$ on $[0, 1]$, prove or disprove

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that $\{f_n\}_{n \geq 1}$ converges pointwise on $[0, 1]$ (8½ marks)

(c) Sketch the first 3 terms of $f_n(x) = x^n - x^{n+1}$ (6marks)

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Question 2 (17 ½ Marks)

(a) State the Uniform Cauchy Criterion theorem (4marks)

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(b) Let $f_n(x) = 5 + x^n \forall x \in [0, 4], n \geq 1$. Find the set S a subset of $[0, 4]$ that well defines the sequence of real valued function. (5marks)

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(c) Show that $f_n(x) = 5 + x^n \forall x \in [0, 1]$ is uniform convergence on $[0, 1]$. (8½ marks)

Question 3 (17 ½ Marks)

(a) State the alternative test of uniform convergence of sequence of functions (3marks)

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(b) Show that $f_n : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f_n(x) = 2$

, $f_n(x) \in \mathbb{R}$ converges on $\mathbb{R}$. (8 ½ marks)

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$\sin f_n(x) f_n(x)$

(c) If $f_n =$ for each $x \in \mathbb{R}$ and $x \in [0, 2\pi]$, then find and show that it does not converge

$\sqrt{x} f_n(x)$

$f_n(x)$

to $\in [0, 2\pi]$. (6 marks)

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Question 4 (17 ½ Marks)

a) When is a bounded function $f: [a, b] \rightarrow \mathbb{R}$ said to be Riemann Integrable? (4marks)

b) Prove that the function $f: [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x^3$ for all $x \in [0, 1]$

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is Riemann Integrable and that $\int_0^1 f(x) dx = 4$ (10 ½ marks)

c) State three (3) basics properties of Riemann Integral (3marks)

Question 5 (17 ½ Marks)

- (a) State and prove Weierstrass M. Test (10marks)
- (b) What is (i) Harmonic series (ii) Alternating Series (iii) P-series (5 ½ marks)
- (c) Give examples of each type in 5(b) (2 marks)

Question 6 (17 ½ Marks)

- a) Define the term Power series and give 2 examples (5marks)
- b) Define radius of convergence. Use your definition to find the radius of convergence of the one of the example of power series you gave in 6(a) (9marks)
- c) State without prove Weierstrass M. Test (3 ½ marks)

Best wishes!!!

Paper 2 of 2 — MTH 225

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COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

SECOND SEMESTER EXAMINATION 2022-2023

Course Code/Title: MTH 225 - Real Analysis IT

Instructions: Attempt any 4 questions. Time Allowed: 2\$ Hours.

Question 1 (17 \$ Marks)

9)- When is a bounded function $f: (a, b] \rightarrow \mathbb{R}$ said to be Riemann Integrable? (4marks)

b) "Prove that the function $f: [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ for all $x \in [0, 1]$ is Riemann Integrable and that $\int_0^1 f(x)dx = \frac{1}{3}$ (10 } marks)

c)~ State three (3) basics properties of Riemann Integral (3marks)

Question 2 (17 # Marks)

(a) Discuss the concept of Pointwise of Convergence of SRVFs (4 marks)

(b) For each integer $n > 1$, $f_n(x) = x^n$ on $[0, 1]$, prove that $\{f_n\}_{n=1}^{\infty}$ converges Pointwise on $[0, 1]$ (8% marks)

(c) Prove or disprove if the Pointwise limit of $\{f_n\}_{n=1}^{\infty}$ is continuous on $[0, 1]$ (5marks)

Question 3 (17 4 Marks)

iform Cauchy Criterion (12 mks)

$f_n(x) = \frac{x^n}{n}$ for $\forall x \in [0, 1]$ converges uniformly on $[0, 1]$ (5+ marks)

interest regarding uniform convergence (3mks)

$n \in \mathbb{N}$ and $x \in [0, 2\pi]$, then find $\lim_{n \rightarrow \infty} f_n(x)$ and show that it does not

> 0 (8 \$ mks)

with examples (i) Harmonic series (ii) Alternating Series (iii) P-

(6 mks)

(3marks)

$f_n(x) = \frac{x^n}{n}$ defined by the series $f_n(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ for $x \in [0, 1]$

(8 \$ marks)

e 2

Sy $(S_n)_{n=1}^{\infty}$ is a sequence of functions on $[0, 1]$ such that $f_n(x) = \frac{x^n}{n}$ (6marks)

gence of a power series? (3marks)

erval of convergence of the series. (8 marks)

at the series converges uniformly on $[0, 1]$ to a

. (6marks)